



UNIVERSIDAD POLITÉCNICA DE SAN LUIS POTOSÍ

Information Technologies Engineering

E-COMMERCE

Group: T56C

**Project Proposal, User Interface and Experience (UI/UX) and
Basic implementation.**

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Business and E-Commerce Proposal

Product or service to be offered:

Our project is an e-commerce website focused on the sale of CDs and vinyl records.

The store will offer regular music products such as albums, special editions, and collectible records. In addition, customers will be able to create personalized CDs and vinyl records that could be used for gifts or personal music production.

The personalization options may include:

- Custom cover artwork.
- Custom labels.
- Personalized packaging.
- Custom messages.
- Custom track lists when the customer owns the music or has permission to use it.
- Special physical editions for independent artists or authorized music.

The main idea is to combine a traditional online music store with a more personal and creative experience.

This means that customers will not only buy music, but they will also be able to create a product that has emotional, artistic, or collectible value.

Target market.

Our main target market includes teenagers, young adults, music fans, collectors, and independent artists.

The store is especially designed for people who enjoy physical music formats and prefer to own a CD or vinyl record instead of only listening to music through streaming platforms.

At the beginning, the business can focus mainly on customers in Mexico, with the possibility of expanding to other countries in the future.

Value proposition.

Our value proposition is to offer physical music products with a high level of personalization. To this day, traditional online music stores usually sell existing albums with fixed designs. Our platform will allow customers to have more control over the final product.

For example, a customer could select a CD, upload personalized artwork, choose a cover design, add a message, and create a unique product. This proposal makes the experience different from a regular music store because the customer becomes "part of the creative process."

Our main value is based on:

- Personalization.
- Collectible products.
- Emotional value.
- Unique gifts.
- Physical music experience.
- Support for independent artists.

The objective is to make music feel more personal and special.

E-commerce business model.

Our main e-commerce business model is an **E-tailer** model, an online retail business that sells products directly to customers through a digital store. Our website will sell physical products such as CDs, vinyl records, special editions, and personalized music products.

The business will mainly operate as **Business-to-Consumer (B2C)** because customers will buy products directly from our online store.

Basic customer journey.

The customer journey describes the steps a person follows when using the website. This journey should be simple, clear, and easy to understand.

1. **Discovery**

The customer discovers the store through social media, online advertisements, recommendations, or visiting the page directly by search engines.

2. **Home Page**

The customer enters the website and sees featured albums, special editions, categories, and personalized products options.

3. **Product Search**

The customer browses the catalog or searches for a specific artist, album, CD, or vinyl record. If it's the case, the user can search for personalization options as well.

4. **Product Details**

The customer opens a product page to see the price, images, description, availability, format, and customization options.

5. **Customization**

If the product is customizable, the customer can select different options such as cover design, label, packaging, or personalized messages. The options must have the required specifications.

6. **Shopping Cart**

The customer adds the product to the shopping cart and can review the selected products.

7. **Checkout**

The customer enters the necessary information and continues with the payment process.

8. **Order Confirmation**

The system confirms the order and provides the customer with the order information.

9. **Delivery**

The product is prepared and shipped to the customer.

Additionally, the customer journey for a personalized product will be:

1. **Select the format**

The customer chooses between CD and vinyl.

2. **Upload audio files**

The customer uploads the songs, recordings, or other audio files that will be included in the product.

3. **Organize the tracks**

The customer can review the uploaded files and arrange their order.

4. **Upload artwork**

The customer can upload an image for the album cover or label.

5. **Add personalization**

The customer can enter a title, message, or other customization details.

6. **Review the configuration**

The website shows a summary of the selected format, files, artwork, and personalization options.

7. **Add to cart**

The customized product is added to the shopping cart with its selected options.

System Design

General system architecture.

Technologies and programming languages to be used.

Initial database design.

Main system components and modules.

Visual Design and UI/UX

Home page.

Product catalog.

Product detail page.

Navigation structure.

Shopping cart interface.

Visual identity, typography, colors, images, and icons.

Basic Functional Prototype

Functional navigation between pages.

The website will include functional navigation between the main pages. The user will be able to move between:

- Home page.
- Product catalog.
- Product detail page.
- Custom CD/Vinyl page.
- Shopping cart.
- Login page.
- Registration page.

A navigation bar will be available at the top of the website.

The navigation bar may include options such as:

- Home
- CDs
- Vinyl
- Custom Products
- Create your own
- Cart
- Login

The objective is to make the website easy to explore and allow the user to move between pages without confusion or extra elements that could affect the user experience.

Product catalog with sample products.

The project requires a catalog with relevant information such as product name, description, price, image, and availability, and the final version must include at least ten different items.

Image	Product	Format	Price	Availability
	Thriller – Michael Jackson	Vinyl	\$699 MXN	Available ↗
	30 - Adele	CD	\$199 MXN	Available ↗
	Back in Black	Vinyl	\$749 MXN	Low Stock ↗
	Custom CD	CD	\$99 MXN	Available ↗

	Custom Vinyl	Vinyl	\$799 MXN	<u>Made to Order</u> ↗
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Each product card will show:

- Product image.
- Album or product name.
- Artist.
- Format.
- Price.
- Availability.
- Button to view more information.

The objective is that the catalog could also include filters by format, artist, price, or category.

Product details.

When a user selects a product, the website will open a product detail page. This page will provide more complete information about the product. It may include:

- Product name.
- Artist's name.
- Album cover.
- Price.
- Format.
- Product description.
- Track list.
- Stock availability.
- Quantity selector.
- Add to cart button.
- Customization options, when available.

For personalized products, additional options may appear. The user should be able to:

- Choose between CD or vinyl.
- Upload audio files.
- Upload a custom cover image.

- Enter an album or project title.
- Add a custom message.
- Choose packaging options.
- Organize the order of the tracks.
- See a summary of the selected customization.

The objective of this page is to give the customer enough information before adding the product to the cart.

Initial database connection or data-management structure.

The prototype will use a relational database to manage the main information of the e-commerce system. The initial structure will include the following tables:

Table	Purpose	Main Fields	Data Type	Connected to
Users	Stores customer and administrator information.	user_id, name, email, password_hash, role	INT, VARCHAR, VARCHAR, VARCHAR, VARCHAR	Shopping_Carts, Custom_Products, Orders
Products	Stores regular CDs, vinyl records, and other products.	product_id, name, artist, format, price, description, image_url	INT, VARCHAR, VARCHAR, VARCHAR, DECIMAL, TEXT, VARCHAR	Categories, Inventory, Cart_Items, Order_Items
Categories	Organizes products into different groups.	category_id, category_name	INT, VARCHAR	Products
Custom_Products	Stores personalized CDs or vinyl records	custom_product_id, user_id, format, title, cover_image, message,	INT, INT, VARCHAR, VARCHAR, VARCHAR,	Users, Custom_Tracks, Cart_Items, Order_Items

	created by users.	packaging_option, price	VARCHAR , TEXT, VARCHAR , DECIMAL	
Custom_Tracks	Stores the audio files included in each personalized product.	track_id, custom_product_id , track_name, file_path, track_order, duration	INT, INT, VARCHAR , VARCHAR , INT, INT	Custom_Products
Shopping_Carts	Stores the active shopping cart of each user.	cart_id, user_id, status, created_at	INT, INT, VARCHAR , DATETIME	Users, Cart_Items
Cart_Items	Stores regular or custom products added to a cart.	cart_item_id, cart_id, product_id, custom_product_id , quantity, unit_price	INT, INT, INT, INT, INT, DECIMAL	Shopping_Carts, Products, Custom_Products
Orders	Stores information about completed purchases.	order_id, user_id, total_amount, order_status, payment_status, created_at	INT, INT, DECIMAL, VARCHAR , VARCHAR , DATETIME	Users, Order_Items, Payments, Shipping
Order_Items	Stores the individual products included in each order.	order_item_id, order_id, product_id, custom_product_id , quantity, unit_price	INT, INT, INT, INT, INT, DECIMAL	Orders, Products, Custom_Products
Inventory	Manages the available quantity of regular products.	inventory_id, product_id, quantity_available, quantity_reserved	INT, INT, INT, INT	Products
Payments	Stores payment	payment_id, order_id, payment_method,	INT, INT, VARCHAR ,	Orders

	transaction information.	payment_status, amount	VARCHAR , DECIMAL	
Shipping	Stores delivery and tracking information.	shipping_id, order_id, shipping_method, tracking_number, shipping_status	INT, INT, VARCHAR , VARCHAR , VARCHAR	Orders

Animations and Interactions

Initial implementation of transitions, hover effects, animations, or micro-interactions.

Interactive elements should contribute to the user experience.